

This week will consist of creating a random number stream and developing a single interactive simulation: a jumping dot game, where you'll apply what you've learned about RxJS to handle user input, state updates, randomness, and rendering.

Random Number Stream

In this exercise, you'll create a stream of random numbers in the range $[-1, 1]$. We already provided helper functions to produce pseudo-random numbers (via a hash), including a helper function `scale` which will help you with scaling down the generated random numbers. The `scale` function takes in a hashed value using `hash` and scales it to the range $[-1, 1]$.

Jumping Dot

In this exercise, you'll build a simple physics-based game where a dot jumps when the spacebar is pressed. The jump is affected by gravity and randomized strength. You'll use observable streams to manage user input, physics updates, and game state in a reactive way.

We've provided a skeleton project and several utility functions to help you focus on building the core RxJS pipelines.